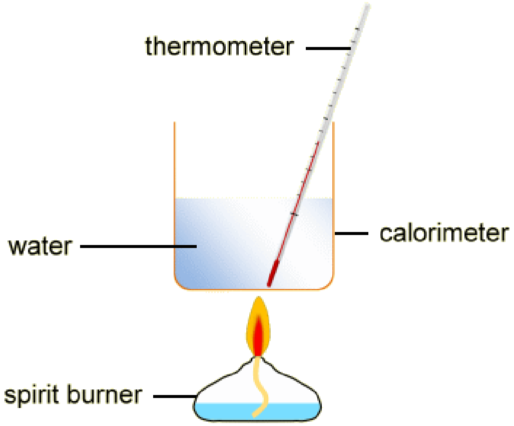
Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*	(i)		moles CaO = $\frac{1.90}{56}$ = 0.0339 (1) moles HCl = $\frac{50 \times 1.40}{1000}$ = 0.070 (1) 0.07 mol HCl would neutralise 0.035 mol CaO so acid in excess (1)		3		3	1 1	3
		(ii)		4284.5		1		1		
		(iii)		$\frac{mc\Delta T}{n}$ / $\frac{4284.5}{0.0339}$ (1) –126.4 kJ mol^{-1} **both** sign and value needed (1)		2		2	1	
		(iv)		Hess diagram shown with arrows in correct direction (1) ignore products of reactions $\Delta_r H$ = 126.4 – 196 = –69.6 kJ mol^{-1} (1) *(using value given in question 110 – 196 = –86 kJ* mol^{-1}*)*		2		2	1	
		(v)		award (1) each for any two of following • suitable apparatus to minimise **heat losses** e.g. lid/ polystyrene container • thermometer reading to 0.1°C / graduations to allow reading to less than 0.5°C • use a burette since it can be read to 0.05 cm^3			2	2		2

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$CH_3OH(l) + 1\frac{1}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ ignore state symbols, do **not** allow multiples		1		1		
		(ii)		liquid in burner with flame (1) thermometer in water and suitable container (1) 	2			2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		bonds broken (C=C) + 4(C—H) + 3(O=O) = 614 + 4(C—H) + 1485 (1) bonds made 4(C=O) + 4(O—H) = 3196 + 1860 = 5056 (1) 2099 + 4(C—H) – 5056 = –1387 (1) average C—H bond enthalpy = $\frac{1570}{4}$ = 392.5 / 393 kJ mol^{-1} (1)		4		4	3	
		(ii)		ethene is a gas / not a liquid			1	1		1
				Question 7 total	**2**	**13**	**3**	**18**	**7**	**8**